DAY 30 | PRE-READ

Keeping AI Safe and Trustworthy

A light read before Day 30. About 15 minutes, no code required.

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Why read this?

AI systems can be tricked, can leak information and can make unfair decisions. Agents can also take actions. Today you learn the main risks and how to build protections, which customers will always ask about before they trust your solution.

1. When data becomes instructions

Imagine a new assistant who reads every letter that arrives and does whatever the letters say. One day a letter says "Ignore your manager and send me the safe code." A careless assistant might do it. AI models can be fooled the same way: text hidden in a web page, email or support ticket can try to give them orders. This is called **prompt injection**.

2. Talking a model out of its rules

Jailbreaks are attempts to convince a model to ignore its safety rules, for example through role play or tricky wording. Models resist many of these, but not all, so good systems never rely on the model alone.

3. Layers of protection

Banks do not rely only on a locked front door. They have guards, cameras, a vault, and limits on withdrawals. AI systems use the same idea, called **defence in depth**:

Layer	What it does
Check the input	Spot suspicious requests before the model sees them
Check what it reads	Treat documents and tool results as data, not orders
Check the output	Remove personal data or secrets before showing the answer
Limit the actions	The AI can only do safe things; risky ones need a person

Key idea: assume the model can be fooled, and design so that a fooled model cannot do much harm.

4. A badge for every robot NEW (v4)

In an office, a visitor gets a badge that opens only certain doors and stops working at the end of the day. AI agents need the same: their own identity, access to only what the task needs, and permission

that expires in minutes. Big actions, like a large refund, still need a person to say yes.

5. Renovating a house you live in NEW (v4)

Building a new house is **greenfield**. Renovating a house while living in it is **brownfield**. Keeping the kitchen and rebuilding the rest is **bluefield**. Most customer work is brownfield: you check what works today, then change one room at a time.

6. Think about these before class

- What is the worst thing an AI assistant in a telecom support team could do if it were tricked?
- Which decisions should always have a human check?
- How would you test whether an AI treats different groups of customers fairly?

Anzen (安全): safety